

Part I

SUMMARY of (In)determinate Forms

Summary

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Some notes on notation for finding limits:

- (a) **The first step in evaluating any limit is to find the form.** This is true, even if the question does not specifically ask for the form.
- (b) To find the form of $\lim_{x \rightarrow a} \frac{f(x)}{g(x)}$, take the limits of the numerator and denominator SEPARATELY. Do not just “plug in $x = a$ ”. “Plugging in” can not occur until after the function is noticed as continuous.
- (c) When writing a form of a limit, we NEVER write $\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{0}{0}$. It is the $=$ that makes this a mathematically incorrect statement. Instead, write $\lim_{x \rightarrow a} \frac{f(x)}{g(x)}$ is of the form $\frac{0}{0}$.
- (d) When evaluating limits, do NOT drop the $\lim_{x \rightarrow a}$ until you have evaluated the limit. For example, the following is INCORRECT:

$$\lim_{x \rightarrow 2} \frac{2x^2 - 4x}{x - 2} = \frac{2x(x - 2)}{x - 2} = 2x = 4$$

Instead, we write:

$$\lim_{x \rightarrow 2} \frac{2x^2 - 4x}{x - 2} = \lim_{x \rightarrow 2} \frac{2x(x - 2)}{x - 2} = \lim_{x \rightarrow 2} 2x = 4$$

Part II

Recitation Questions

Problem 1

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Problem 1

Problem 1 Which of the following limits are of the form $\frac{0}{0}$?

(a) $\lim_{x \rightarrow 7} \frac{\sin(x-7)}{|x-7|}$

(b) $\lim_{x \rightarrow \frac{\pi}{2}} \frac{\sin(x)}{x}$

(c) $\lim_{x \rightarrow \frac{\pi}{2}} \frac{|\cos(x)|}{x}$

(d) $\lim_{x \rightarrow \frac{\pi}{2}} \frac{\pi - 2x}{\cos(x)}$

(e) $\lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 4}$

(f) $\lim_{x \rightarrow 4} \frac{x^2 - 4}{x - 4}$

(g) $\lim_{x \rightarrow 4} \frac{x^2 - 16}{x - 4}$

(h) $\lim_{x \rightarrow 3} \frac{x - 4}{\sqrt{25 - x^2} - 4}$

(i) $\lim_{x \rightarrow 3} \frac{x - 3}{\sqrt{25 - x^2} - 4}$

Problem 2

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Problem 2

Problem 2 State the form of the limit and then evaluate the limit.

(a) $\lim_{x \rightarrow 6} \frac{4x^2 - 144}{x - 6}$

(b) $\lim_{x \rightarrow 6} \frac{x - 6}{\sqrt{2x - 8} - 2}$

(c) $\lim_{x \rightarrow 2} \frac{(3x - 2)^2 - 16}{x - 2}$

(d) $\lim_{x \rightarrow 6} \frac{\frac{x}{x-2} - \frac{3}{2}}{x - 6}$

(e) $\lim_{x \rightarrow 1} \frac{\sqrt{5x - 2} - \sqrt{3}}{x - 1}$

Problem 3

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Problem 3

Problem 3 Evaluate the following limits.

(a) $\lim_{x \rightarrow \frac{\pi}{2}} \frac{\cos(x)}{x}$

(b) $\lim_{x \rightarrow 3} \frac{\sqrt{25 - x^2} - 4}{x - 4}$

(c) $\lim_{x \rightarrow 3} \frac{\sqrt{25 - x^2} - 4}{x - 3}$

Problem 4

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Problem 4

Problem 4 Two polynomials, h and g , are given

$$h(x) = \frac{x^2 - 4}{4}$$

$$g(x) = x - 2$$

State the form of the limit, evaluate the limit, or state that it does not exist.
Justify your answer.

(a) $\lim_{x \rightarrow 2} \frac{h(x)}{g(x)}$

(b) $\lim_{x \rightarrow 3} \frac{g(x) - g(3)}{x - 3}$

(c) $\lim_{x \rightarrow 4} \frac{h(x) - h(4)}{x - 4}$

Problem 5

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Problem 5

Problem 5 Evaluate each of the following limits. Possible answers include a number, $+\infty$, $-\infty$ and “Does Not Exist” (DNE). Make sure to state the form of the limit. Justify your answer.

(a) $\lim_{x \rightarrow 3^-} \frac{x^2 - 3}{x^2 - x - 6}$

(b) $\lim_{x \rightarrow 5^+} \frac{x^2 + 6}{x^2 - 3x - 10}$

(c) $\lim_{x \rightarrow 1} \frac{4 - x}{x^2 - 2x + 1}$

(d) $\lim_{x \rightarrow 2} \frac{-e^x}{(2 - x)^3}$

(e) $\lim_{x \rightarrow 1} \frac{\sin x}{\sqrt{2 - x^2} - 1}$

Problem 6

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Problem 6

Problem 6 A piecewise defined function f is given by

$$f(x) = \begin{cases} \frac{2x-3}{x-2} & \text{if } x < 2 \\ \frac{x^2-5x+6}{x^2-4} & \text{if } x > 2 \end{cases}$$

Determine the form of the limit, then find the limit.

(a) $\lim_{x \rightarrow 2^+} f(x)$

(b) $\lim_{x \rightarrow 2^-} f(x)$

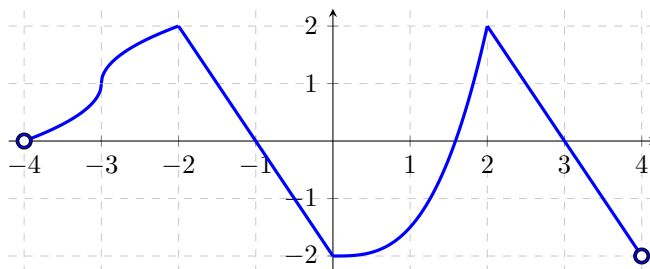
(c) $\lim_{x \rightarrow 2} f(x)$

Problem 7

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Problem 7

Problem 7 The graph of a function $g(x)$ with domain $(-4, 4)$ is given in the figure. This portions of this graph in the intervals $(-2, 0)$ and $(2, 4)$ are straight lines. The portion on the interval $(0, 2)$ is not parabolic.



For each of the limits below, give the form of the limit, then evaluate the limit.

(a) $\lim_{x \rightarrow -3} \frac{g(x) + \sin\left(\frac{\pi}{4}x\right)}{x\sqrt{x+4}}.$

(b) $\lim_{x \rightarrow 0^+} \frac{e^x}{|g(x) + 2|}.$

(c) $\lim_{x \rightarrow 3} \frac{x^2 + 2x - 15}{x + g(x) - 3}$